

Lecture Notes

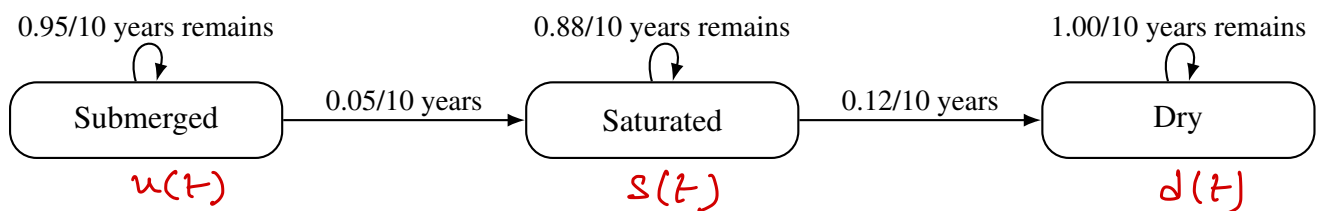
on Vector Dynamics (Cont...)
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Initial condition

$$\begin{pmatrix} u(0) \\ s(0) \\ d(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

✎ **Example:** Suppose that we are looking at a total area of 100 hectares, which is initially all in **submerged wetland**. Every 10 years, 5% of submerged wetlands become **saturated** wetlands, 12% of saturated wetlands become **dry**, and all hectares that are dry remain dry. After 10 years, how much of the land is submerged, how much is saturated, and how much is dry? After 20 years?

Figure 1: Flow Diagram for the Transfer Matrix



Vector Solution:

let $u(t)$, $s(t)$ and $d(t)$ denote the submerged, saturated and dry areas respectively, where t denotes the time, with time step 10 years.

Given the initial condition $\begin{pmatrix} u(0) \\ s(0) \\ d(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

let $\vec{v}(t) = \begin{pmatrix} u(t) \\ s(t) \\ d(t) \end{pmatrix}$, where $\Rightarrow u(t+1) = u(t) - 0.05u(t) = 0.95u(t)$

$$u(t) \Big|_{t=10} = u(10) = 1 - (0.05 \times 1) = 0.95$$

$$s(t) \Big|_{t=10} = s(10) = 0 + (0.05 \times 1) - (0 \times 0.12) = 0.05$$

$$s(1) = s(0) + (0.05 \times u(0)) - (0.12 \times s(0))$$

$$d(t) \Big|_{t=10} = d(10) = 0 + (0 \times 0.12) - 0 = 0$$

$$d(1) = d(0) + 0.12s(0) - d(0) = d(t) + 0.12s(t)$$

$$\text{Thus } \vec{v}(10) = \begin{pmatrix} u(10) \\ s(10) \\ d(10) \end{pmatrix} = \begin{pmatrix} 0.95 \\ 0.05 \\ 0 \end{pmatrix}.$$

So, after 10 years the required land distribution is

$$\text{given by } 100 \vec{v}(10) = \begin{pmatrix} 100 \times 0.95 \text{ hec} \\ 100 \times 0.05 \text{ hec} \\ 100 \times 0 \text{ hec} \end{pmatrix} = \begin{pmatrix} 95 \\ 5 \\ 0 \end{pmatrix} \text{ hec.}$$

$$\text{Thus, } \vec{v}(t+1) = \begin{pmatrix} u(t+1) \\ s(t+1) \\ d(t+1) \end{pmatrix} = \begin{pmatrix} 0.95 u(t) \\ 0.88 s(t) + 0.05 u(t) \\ d(t) + 0.12 s(t) \end{pmatrix}$$

$$\begin{aligned} \text{So, } \vec{v}(20) &= \vec{v} / \text{after 20 years} = \begin{pmatrix} 0.95 u(10) \\ 0.88 s(10) + 0.05 u(10) \\ d(10) + 0.12 s(10) \end{pmatrix} \\ &= \begin{pmatrix} 0.9025 \\ 0.0915 \\ 0.006 \end{pmatrix} \end{aligned}$$

Hence the required land distribution is given by

$$100 \vec{v}(20) = \begin{pmatrix} 90.25 \\ 9.15 \\ 0.6 \end{pmatrix} \text{ hec.}$$

Note. This is the approach where we solve the problem using **vectors**. Next, we will see how to solve the same problem using **matrices**. Moreover, observe that we are essentially creating a **sequence of vectors** for each time step.

Matrix Solution:

$$\vec{v}(t+1) = \begin{pmatrix} u(t+1) \\ s(t+1) \\ d(t+1) \end{pmatrix} = \begin{pmatrix} 0.95u(t) \\ 0.88s(t) + 0.05u(t) \\ d(t) + 0.12s(t) \end{pmatrix}$$

$$\Rightarrow \vec{v}(t+1) = \underbrace{\begin{pmatrix} 0.95 & 0 & 0 \\ 0.05 & 0.88 & 0 \\ 0 & 0.12 & 1 \end{pmatrix}}_{A \text{ (Transfer matrix)}} \begin{pmatrix} u(t) \\ s(t) \\ d(t) \end{pmatrix}$$

$$\text{Thus, } \vec{v}(10) = A \vec{v}(0)$$

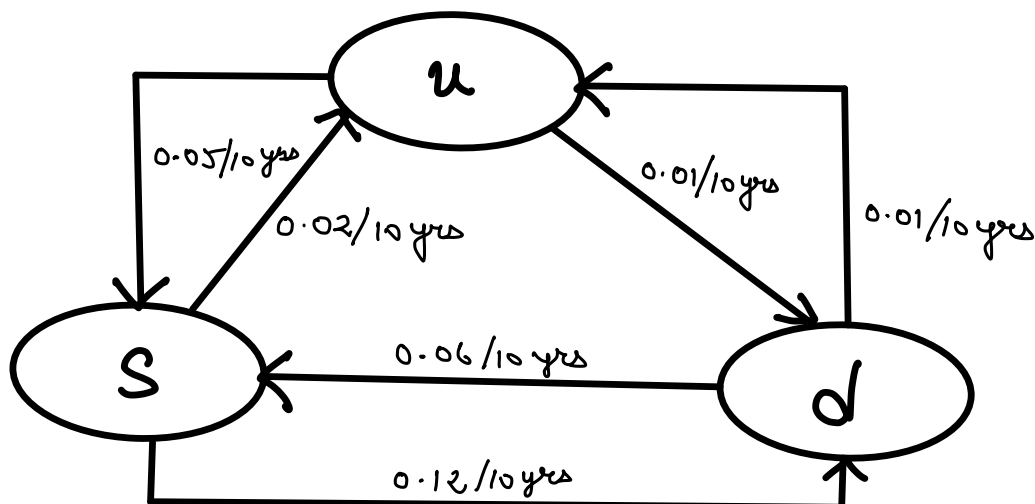
$$\vec{v}(20) = A \vec{v}(10) = A(A \vec{v}(0)) = A^2 \vec{v}(0)$$

$$= \begin{pmatrix} 0.95 & 0 & 0 \\ 0.05 & 0.88 & 0 \\ 0 & 0.12 & 1 \end{pmatrix} \begin{pmatrix} 0.95 & 0 & 0 \\ 0.05 & 0.88 & 0 \\ 0 & 0.12 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.9025 & 0 & 0 \\ 0.0915 & 0.7744 & 0 \\ 0.006 & 0.2256 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.9025 \\ 0.0915 \\ 0.006 \end{pmatrix} \quad (\text{same answer as the previous method})$$

More complex flow diagram.



moving from →

	$u(t)$	$S(t)$	$d(t)$
$u(t+1)$	a_{11}	a_{12}	a_{13}
---	---	---	---
$S(t+1)$	a_{21}	a_{22}	a_{23}
---	---	---	---
$d(t+1)$	a_{31}	a_{32}	a_{33}

← moving into

$$a_{11} = 1 - 0.05 - 0.01 = 0.94.$$

$$a_{12} = 0.02$$

$$a_{13} = 0.01$$

$$a_{21} = 0.05.$$

$$a_{22} = 1 - 0.02 - 0.12 = 0.86$$

$$a_{23} = 0.06$$

$$a_{31} = 0.01$$

$$a_{32} = 0.12$$

$$a_{33} = 1 - 0.01 - 0.06 = 0.93$$

$$\vec{v}(t+1) = \underbrace{\begin{pmatrix} 0.94 & 0.02 & 0.01 \\ 0.05 & 0.86 & 0.06 \\ 0.01 & 0.12 & 0.93 \end{pmatrix}}_A \vec{v}(t)$$

$$\vec{v}(t+1) = A \vec{v}(t)$$

$$\vec{v}(1) = A \vec{v}(0)$$

$$\vec{v}(2) = A \vec{v}(1) = A(A \vec{v}(0)) = A^2 \vec{v}(0)$$

...

$$\vec{v}(n) = A^n \vec{v}(0)$$

Thus for this more complex example, the land distribution after 20 years is given by [where initial condition is $\vec{v}(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$]

$$\vec{v}(20) = \begin{pmatrix} 0.94 & 0.02 & 0.01 \\ 0.05 & 0.86 & 0.06 \\ 0.01 & 0.12 & 0.93 \end{pmatrix}^{20} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.94 & 0.02 & 0.01 \\ 0.05 & 0.86 & 0.06 \\ 0.01 & 0.12 & 0.93 \end{pmatrix} \begin{pmatrix} 0.94 & 0.02 & 0.01 \\ 0.05 & 0.86 & 0.06 \\ 0.01 & 0.12 & 0.93 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.8847 & 0.0372 & 0.0199 \\ 0.0906 & 0.7478 & 0.1079 \\ 0.0247 & 0.215 & 0.8722 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.8847 \\ 0.0906 \\ 0.0247 \end{pmatrix}$$